

1 An architect firm specialises in designing skyscrapers.

The firm uses high end computers with high performance CPUs, GPUs and large amounts of RAM.

i. Give one use the firm might have for GPUs.

----- [1]

ii. Describe what is meant by the term 'RAM'.

----- [2]

iii. State **one** characteristic a high performance CPU might have.

----- [1]

2 Define the term 'multicore processor'.

----- [2]

- 3 * A charity has several desktop computers in their office that use a CISC processor. They are considering buying mobile devices for their staff to use when they are not in the office.

Discuss whether these mobile devices should use the same CISC processors that are used in their desktop computers or if they should use a RISC processor instead.

You should include the following in your answer:

- the difference between each processor type
- the suitability of each processor type for mobile devices.

This image shows a blank sheet of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no other markings, text, or illustrations on the page.

This image shows a full page of a document template. It consists of approximately 28 horizontal dashed lines spaced evenly down the page, providing a guide for handwriting or typing. The lines are thin and black against a plain white background.

- 4 OCRSystems are designing a new CPU for a computer system that will be used for video rendering. Part of the video rendering process is when the video is exported. This is when the computer combines all of the separate video elements together to form the final video.

An important design consideration is whether OCRSystems use a CISC processor type or a RISC processor type.

Describe **one** difference between a CISC processor and a RISC processor.

[2]

- 5 CPUs can be designed to take a Complex Instruction Set Computer (CISC) or a Reduced Instruction Set Computer (RISC) approach.

Describe **one** difference between CISC and RISC.

[2]

6 A company produces digital photo frames (i.e. photo frames that display digital photographs).

The photo frame has a RISC processor.

Describe **one** advantage to the company of using a RISC processor, rather than a CISC processor.

[2]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1		i	To render models of proposed buildings. (1) Run CAD software. (1) Run modelling calculations. (1) Any example sensible to scenario. (1) (Max 1)	1 (AO2.1)	Examiner's Comments Some candidates did not apply the use of GPU to the scenario. Those who did, generally gained full marks for this question.
		ii	Random Access Memory (1) A form of primary memory (1) Used to hold data and / or programs <u>in use</u> (1) Volatile / Loses its contents when power is lost. (1) (Max 2)	2 (AO1.1)	
		iii	Multiple Cores (1) High / Fast Clock Speed (1) Ability to use pipelining (1) Large Cache (1) (Max 1)	1 (AO1.1)	Accept concurrency / parallel processing for pipelining
			Total	4	
2			– more than one processing unit in a (single) processor – ...which can independently process instructions at the same time.	2 AO1.1	Examiner's Comments Too many candidates defined a multi core processor as 'a processor with multiple cores' this is not creditworthy at this level of study. Candidates must demonstrate their understanding of a core being a processing unit within a processor.
			Total	2	
3			Mark Band 3–High Level (9-12 marks) The candidate demonstrates a thorough knowledge and understanding of both CISC and RISC. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation.	12 (AO1.1) (2), (AO1.2) (2), (AO2.1) (3), (AO3.3). (5)	AO1 CISC is a complex instruction set. The traditional approach to processor design. Lots of instructions available although some instructions in CISC will rarely get used. RISC is a reduced instruction set. A smaller number of instructions available, several instructions can be combined to perform the same tasks as CISC processors. RISC instructions are used regularly.

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
	The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2-Mid Level (5-8 marks) The candidate demonstrates reasonable knowledge and understanding of CISC and/or RISC; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-4 marks) The candidate demonstrates a basic knowledge of CISC or RISC; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		RISC has fewer transistors/less complex circuitry whereas CISC integrated circuits are more expensive/complicated. RISC instructions take one cycle whereas CISC may take several. RISC can only do complex things by combining multiple instructions whereas CISC is done in one line. Compilers for RISC need to be more complex than compilers for CISC AO2 CISC processors would run the same software as the desktop machines. Would be less power efficient and require larger battery and cooling mechanisms. More expensive to purchase. RISC processor requires software to be written specifically for it (cannot use CISC instructions). More power efficient and so requires less/no cooling and smaller battery/longer battery life. RISC devices may require greater RAM as programs tend to be larger than their CISC equivalents. AO3 Mobile use of CISC would save money on software and increase compatibility but cost more to purchase and be physically larger (heat sink/larger battery) and/or have a shorter battery life. RISC would require investment in software but be cheaper to purchase and give a better performance out of the office (lighter/longer battery life). Some compatibility issues may be reduced with emulators and translators. Examiner's Comments Many candidates were able to discuss the difference in reduced or complex instruction sets and gave some discussion of the increase in hardware requirements for CISC. Few talked about the software differences, and some assumed the charity would need to be programming the devices which was not relevant to the question.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	12	
4			• CISC has a larger instruction set • RISC has a smaller instruction set • CISC is difficult to pipeline • RISC is easier to pipeline • CISC tends to have more addressing modes • RISC tends to have fewer addressing modes • CISC instructions may take multiple clock cycles to execute • RISC instructions take one clock cycle to execute • CISC has complex circuitry/more transistors. • RISC has less complex/simple circuitry • CISC uses less RAM • RISC uses more RAM	AO1.2 (2)	Accept any other valid points Mark in pairs <u>Examiner's Comments</u> This question was generally answered well with many candidates describing the number of clock cycles required to execute instructions for RISC and CISC. Some candidates were not given the marks because they did not use the correct terminology.
			Total	2	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5			1 mark per bullet up to a maximum of 2 marks, e.g: • Some instructions in CISC will rarely get used ... • ... In RISC instructions are used regularly • In assembly for CISC, a statement that takes one mnemonic ... • ... (may) take multiple mnemonics in RISC • Compilers for RISC need to be more complex ... • ... than compilers for CISC • CISC architecture has complex circuitry and is therefore more expensive to manufacture... • ...RISC architecture has simple circuitry minimising manufacture cost.	2 AO1.1 (1) AO1.2 (1)	Accept More than one clock cycle for each instruction in CISCone clock cycle for each instruction in RISC For BP's 3 and 4
			Total	2	
6			<i>advantage</i> - Costs less to design/produce. <i>because</i> - Requires less cooling to be built in. - (If battery powered) can run off smaller battery. - Has fewer instructions than other (CISC) processors. - Simpler (circuit/hardware) design/manufacture. - Fewer transistors. (1 per -, max 2, 1 max from 'because' group)	2 AO2.1 (2)	<u>Examiner's Comments</u> Some candidates did not gain credit because the advantage stated was not specific to the company, as the question asked. Candidates should be reminded to read the question carefully.
			Total	2	